

Reviewer Pack — Fourier Coefficient Spread and ACC⁰ Circuit Lower Bounds

Entry 90abe309451a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Fourier Coefficient Spread and ACC⁰ Circuit Lower Bounds

Entry ID: 90abe309451a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-12 05:17:09 UTC

1. Conjecture

Field A (mathematical branch): Fourier Analysis on Finite Groups **Field B** (complexity object): ACC⁰ Circuit Size

Statement:

For any boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, if the sum of absolute Fourier coefficients $\sum |\hat{f}(S)| \geq 2^{\{n/2\}} \sqrt{n}$, then f cannot be computed by an ACC⁰ circuit of size $2^{\{n/2\}}$.

Rationale (proposer's reasoning):

The spread of Fourier coefficients reflects correlation with parity functions. ACC⁰ circuits, limited to bounded-depth polynomial-size gates, struggle to capture such correlations, making large Fourier sums a barrier to efficient computation.

Taxonomy category: FOURIER_ANALYTIC (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b72019fcb3187aa9

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	UNCERTAIN	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def fast_walsh_hadamard_transform(f):
    n = len(f)
    if n == 1:
        return f
    even = fast_walsh_hadamard_transform([f[i] for i in range(0, n, 2)])
    odd = fast_walsh_hadamard_transform([f[i] for i in range(1, n, 2)])
    result = [0] * n
    for k in range(n // 2):
        result[k] = even[k] + odd[k]
        result[k + n // 2] = even[k] - odd[k]
    return result

def generate_random_boolean_function(n):
    return [random.randint(0, 1) for _ in range(2**n)]

def compute_fourier_coefficients(f):
    n = int(math.log2(len(f)))
    f_transformed = fast_walsh_hadamard_transform(f)
    fourier_coeffs = [abs(f_transformed[k] / len(f)) for k in range(n)]
    return sum(fourier_coeffs)

def is_in_acc0_circuit_size(f):
    # Placeholder function to check if a boolean function can be computed by an ACC0 circuit
    # This is a dummy implementation and should be replaced with actual logic
    return False

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 10
    f = generate_random_boolean_function(n)
    fourier_sum = compute_fourier_coefficients(f)
    conjecture_holds = fourier_sum < 2**(n/2) * math.sqrt(n)
    counterexample = "" if conjecture_holds else "mapping_undefined"
    return {
        "metric_name": "Fourier Coefficient Sum",
        "metric_value": fourier_sum,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
```

```

import sys
seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]
results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value)**2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next(s for s, r in zip(seeds, results) if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
else:
    print(f"RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

tric_name': 'Fourier Coefficient Sum', 'metric_value': 0.64453125, 'instances_tested': 1, 'conjecture_holds': 1
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.626953125, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.634765625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.546875, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.56640625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.5546875, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.59765625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.658203125, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.625, 'instances_tested': 1, 'conjecture_holds': 1}
TRIAL: {'metric_name': 'Fourier Coefficient Sum', 'metric_value': 0.666015625, 'instances_tested': 1, 'conjecture_holds': 1}
RESULT: SUPPORTED mean=0.6087239583333334 std=0.03415580763030604 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

{ "critic_verdict": "CHALLENGE", "reasoning": "The test uses $n=1$ (implied by metric values <1) where $2^{\{n/2\}\sqrt{n}} = 2^{\{0.5\}\sqrt{1}}$ " }

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test used $n=1$ where the conjecture's threshold ($2^{\{n/2\}\sqrt{n}}$) is ~ 1.414 , but all metric values (<1) never reached this threshold. Thus, the conjecture's condition was never

tested. | next: Test with $n \geq 2$ where $2^{\{n/2\}} \sqrt{n} \geq 2$ to validate the conjecture’s actual condition

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	88743
2	preregistration	ollama_remote	qwen3:8b	0	0	25718
3	novelty	ollama_remote	qwen3:8b	0	0	20767
4	novelty	ollama_remote	qwen3:8b	0	0	15772
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9988
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8025
7	critic	ollama_remote	qwen3:8b	0	0	30290
8	judge	ollama_remote	qwen3:8b	0	0	20768

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 220071 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/90abe309451a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/90abe309451a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/90abe309451a.tar.gz` (if generated)